

MERIDIAN TRUST & ASSET SERVICING, N.A.

Central AI & Model Inventory · Global Custody Operations

CAIMI-INT-02 / UC-2103

AI Use-Case Intake & Registration

Custody Settlement Failure Resolution Synthesizer (CS-FRS) v1.2

Document number	CAIMI-INT-02 / UC-2103
Version	1.2
Classification	Internal
Document owner	Managing Director, Global Custody Operations
Effective	Current cycle
Review cycle	Annual
Approved by	Model Risk Management — concurred

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Document Control

Form	CAIMI-INT-02 (supersedes MRM-INT-01 for systems in scope of STD-AI-002)
CAIMI System ID	AI-REG-2026-0117
Use-Case ID	UC-2103
Submitted by	First Line of Defense
Reviewed by	Second Line of Defense — Model Risk Management
Status	Validation complete — awaiting production release authorisation

Registration precedes validation. Validation precedes production. No system may connect to a production data source until this record is complete and 2LOD concurrence is recorded at Section G.

Section A — Identification

Field	Entry
CAIMI System ID	AI-REG-2026-0117
Use-Case ID	UC-2103
Application Name	Custody Settlement Failure Resolution Synthesizer (CS-FRS) v1.2
Sponsoring Division	Global_Custody
Business Process Owner (1LOD)	Managing Director, Global Custody Operations
Technical Lead (1LOD)	Solutions Architect, Asset Servicing Technology
Second Line Reviewer (2LOD)	Model Risk Officer, AI Systems
Submission date	Current cycle
Development status	Validation complete — awaiting production release authorisation

Section B — Functional Design

Field	Entry
Intended Purpose	Drafts operational resolution narratives for failed securities settlements. On receipt of an inbound <code>MT548</code> fail advice, the system parses the status qualifier, matches the trade against Omgeo CTM confirmation records and internal depot instructions, identifies the mismatched field, and drafts a resolution ticket citing the applicable depository clearing procedure.
Restrictions on Use	Draft-only. The system may not transmit settlement instructions, may not amend a standing settlement instruction, and may not release assets. Every output is a proposal in a ServiceNow queue awaiting clerk authorisation.
Technology Typology	<code>RAG_Pipeline</code>
Base Model Supplier	Anthropic, via Amazon Bedrock
Foundation Model	FM-0007 — registered shared component; see <code>foundation-model-registration-aibom.md</code>
Integration Endpoints	SWIFT Alliance (inbound MT548 read-only), Omgeo CTM (read-only), internal depot instruction store (read-only), ServiceNow FSO (write — ticket creation only)
Decision made by the system	None. The system proposes a resolution hypothesis. The settlement clerk verifies the correction against the authorised SSI directory and authorises transmission.

Operational workflow

Stage	Input	Mechanism	Resulting state
1 Ingestion	Inbound <code>MT548</code> from clearing broker	Extracts status qualifiers, e.g. <code>:24B::PENF//LACK</code>	Structured payload
2 Matching	Payload + trade ticket ID	Queries CTM confirmations and depot instructions	Mismatched field identified
3 Synthesis	Mismatch + clearing procedure corpus	Drafts resolution narrative with citation to the governing procedure	Draft ticket in ServiceNow queue
4 Adjudication	Draft ticket	Settlement clerk verifies against the authorised SSI directory	Clerk authorises or rejects
5 Transmission	Authorised ticket	Operations interface generates <code>MT542</code> amendment	Trade re-queued for matching

Stage 4 is a hard gate. The system has no path to stage 5. An unauthorised ticket expires in the queue.

Section C — Data Classification

Field	Entry
Data Sensitivity Tier	Internal_Restricted
Data Sources	BIC codes, LEIs, CUSIP/ISIN identifiers, nominal share counts, cash settlement amounts, depository error codes, internal clearing procedure manuals
PII Consumption Flag	False — client legal names, account numbers, and natural-person identifiers are stripped at the network layer before ingestion. Filtering is enforced by the integration gateway, not by the application.
MNPI Handling Flag	False
GLBA Covered Data	False
Residency Boundary	US_East_Dedicated_Enclave

Section D — Fiduciary Scope

Field	Entry
Legal Capacity	Agency_Custodial — pure custody and clearing agency
Discretion exercised	None. The Bank holds no investment discretion over the affected assets.
Impartiality Impact Flag	False — the system cannot affect allocation between beneficiary classes
12 CFR 9.6(c) Review Role	None
ERISA Advice Flag	False
Regulatory exposure	Operational only. No Part 9 fiduciary duty, no ERISA fiduciary duty, no UPIA duty attaches to this workflow.

Section E — Human Oversight and Reversibility

Field	Entry
Automation Level	Strict_Human_In_The_Loop — mandatory clerk authorisation on every output, no exception path
Execution Reversibility	Fully_Reversible_T0
Rollback Procedure Ref	SOP-CUST-114 — settlement instruction amendment and recall
Failure consequence	An incorrect resolution hypothesis is rejected either by the clerk at stage 4 or by the depository matching engine at stage 5. Maximum realistic impact is localised operational delay within the trading day. No capital loss pathway has been identified.

Section F — Supply Chain

Field	Entry
Hosting Environment	Bank_Private_VPC
Zero Data Retention SLA	True — contractually confirmed for the operational-identifier data class processed by this system
AIBOM Identifier	CDX-MTAS-0117 (CycloneDX ML-BOM v1.7)
Third-party assessment	Completed under the interagency third-party guidance; see shared foundation-model registration

Section G — Tiering and Sign-Off

Inherent risk score — computed under STD-AI-002 § 5:

Dimension	Weight	Score	Weighted	Basis
Fiduciary & legal exposure	0.25	1	0.25	Back-office, non-fiduciary agency function
Data sensitivity & privacy	0.20	1	0.20	Operational identifiers only; PII filtered upstream
Autonomy & human oversight	0.20	1	0.20	Strict human-in-the-loop, no exception path
Operational reversibility	0.15	1	0.15	Fully reversible within trading day
Model complexity	0.10	5	0.50	Generative LLM with retrieval
Financial & volumetric exposure	0.10	3	0.30	Moderate aggregate settlement volume
Inherent Risk Score			1.60	

Field	Entry
Determined Risk Tier	Tier_3_Low (score 1.60 < 2.50)
Mandatory override applicable?	No — not fiduciary capacity, no impartiality impact, no beneficiary PII, not irreversible
1LOD Owner Signature	Managing Director, Global Custody Operations — signed, current cycle
2LOD MRM Concurrence	Model Risk Officer, AI Systems — concurred . Tier 3 determination accepted.

Section H — Attached Evidence

- `evidence-data-governance.md` — data boundary, network-layer PII filtering, residency attestation
- `evidence-human-oversight-raci.md` — the stage-4 gate, RACI, and clerk authorisation controls
- `evidence-monitoring-plan.md` — KRIs, thresholds, escalation, review cadence
- `evidence-rag-grounding.md` — source allow-list, golden dataset, context precision / recall / faithfulness results
- `evidence-validation-memo.md` — 2LOD streamlined verification against gold-standard manual resolutions
- Shared: `enterprise-ai-governance-standard.md`, `foundation-model-registration-aibom.md`, `fiduciary-terms-glossary.md`

Section I — Owner Attestation

The use-case owner attests that the description above is accurate, that the system operates within the stated restrictions, that no natural-person identifiers reach the model, and that the stage-4

authorisation gate cannot be bypassed by configuration. Signed: Managing Director, Global Custody Operations.

SYNTHETIC DEMONSTRATION DOCUMENT. Meridian Trust & Asset Servicing, N.A. is a fictional institution. This registration and all evidence referenced by it were authored for a SAS product demonstration.